

◆ 1. Basic Setup (common for all optimizers)

```
import torch
import torch.nn as nn
import torch.optim as optim

model = nn.Linear(10, 1) # example model
```

👉 General pattern:

```
optimizer = optim.OptimizerName(model.parameters(), lr=learning_rate)
```

◆ 2. Core Optimizers

✅ 1. SGD (with optional Momentum)

```
optimizer = optim.SGD(model.parameters(), lr=0.01)
```

With Momentum:

```
optimizer = optim.SGD(model.parameters(), lr=0.01, momentum=0.9)
```

With Nesterov:

```
optimizer = optim.SGD(model.parameters(), lr=0.01, momentum=0.9, nesterov=True)
```

✅ 2. Adam (Most Used)

```
optimizer = optim.Adam(model.parameters(), lr=0.001)
```

Custom:

```
optimizer = optim.Adam(model.parameters(), lr=0.001, betas=(0.9, 0.999))
```

✅ 3. RMSprop

```
optimizer = optim.RMSprop(model.parameters(), lr=0.001)
```

✓ 4. Adagrad

```
optimizer = optim.Adagrad(model.parameters(), lr=0.01)
```

✓ 5. Adadelta

```
optimizer = optim.Adadelta(model.parameters(), lr=1.0)
```

✓ 6. AdamW (Better version of Adam)

```
optimizer = optim.AdamW(model.parameters(), lr=0.001)
```

◆ 3. Less Common but Important

✓ 7. Adamax (variant of Adam)

```
optimizer = optim.Adamax(model.parameters(), lr=0.002)
```

✓ 8. ASGD (Averaged SGD)

```
optimizer = optim.ASGD(model.parameters(), lr=0.01)
```

✓ 9. LBFGS (Second-order optimizer)

```
optimizer = optim.LBFGS(model.parameters(), lr=0.1)
```

⚠ Special usage (needs closure):

```
def closure():
    optimizer.zero_grad()
    loss = model(input).sum()
    loss.backward()
    return loss
```

```
optimizer.step(closure)
```

✓ 10. SparseAdam (for sparse data like embeddings)

```
optimizer = optim.SparseAdam(model.parameters(), lr=0.001)
```

✓ 11. Rprop (Resilient backprop)

```
optimizer = optim.Rprop(model.parameters(), lr=0.01)
```

◆ 4. Training Loop Example (important)

```
for epoch in range(epochs):
    optimizer.zero_grad()      # reset gradients
    outputs = model(X)
    loss = loss_fn(outputs, y)
    loss.backward()            # compute gradients
    optimizer.step()            # update weights
```
